Homework

write in $\frac{p}{q}$ form

1.) $12.5\overline{9}$

2.) $18.5\overline{7}$

3.) $0.00\overline{32}$

4.) $0.253253...$

5.) $1.\overline{2} + 3.7\overline{6}$

6.) $0.\overline{6} + 0.\overline{7} + 0.4\overline{7}$

Q. $13.1\overline{45}$

$\Rightarrow \frac{13145 - 131}{990} \Rightarrow \frac{13014}{990}$

Q. $1.12\overline{3}$

$\Rightarrow \frac{1123 - 112}{900} = \frac{1011}{900}$

Decimal Expansion of Real numbers

- Rational no.
  - Terminating
    - → Prime factorisation of denominator contain 2 or 5 or both as factor
  - non-terminating Repeating
    - → Prime factorisation contain factors other than (3, 7, 11 etc) than 2 or 5 or both.
- Irrational no.

Note → $\frac{p}{q}$ → Rational no.

⇒ ** p and q must be co-prime.

Eg → $\frac{6}{15} = \frac{6}{3 \times 5} = \frac{2}{5} = 0.5$ (terminating)

- $\frac{1}{3} = 0.33...$ (non-terminating and Repeating)
- $\frac{1}{2} = 0.5$ (terminating)
- $\frac{1}{6} = 0.166...$ (non-terminating and Repeating)
- $\frac{1}{4} = 0.25$ terminating)
- $\frac{1}{5} = 0.2$ (terminating)

Note :- In the case of terminating decimal expansion, the remainder comes 0.

- In the case of non-terminating, repeating decimal expansion, the remainder never comes 0.

Ex-1.3

Q1. a) $\frac{36}{100} = 0.36$

- terminating decimal expansion.

b) $\frac{1}{11} = 0.\overline{09}$

- non-terminating but repeating decimal expansion.

c) $4\frac{1}{8} = \frac{33}{8} = 4.125$

- terminating decimal expansion

d.) $\frac{3}{13} = 0.\overline{230769}$

- non terminating repeating decimal expansion

c) $\frac{2}{11} = 0.\overline{18}$

- non terminating repeating decimal expansion

f.) $\frac{329}{400} = 0.1225$

- terminating decimal expansion.

Q2.

sol: $\frac{1}{7} = 0.\overline{142857}$

$\cdot\frac{2}{7} = 2 \times 0.\overline{142857}$
$= 0.\overline{285714}$

$\cdot\frac{3}{7} = 3 \times 0.\overline{142857}$
$= 0.\overline{428571}$

$\cdot\frac{4}{7} = 4 \times 0.\overline{142857}$
$= 0.\overline{571428}$

$\cdot\frac{5}{7} = 5 \times 0.\overline{142857}$
$= 0.\overline{714285}$

$\cdot\frac{6}{7} = 6 \times 0.\overline{142857}$
$= 0.\overline{857142}$

Q5.

sol: $\frac{1}{17} = 0.\overline{0588235294117647}$

```
17 | 100 (0.0588235294117
     85     647058...
     150
     136
      140
      136
       40
       34
       60
       51
       90
       85
       50
       34
       160
       153
        70
        68
        20
        17
        30
        17
        130
        119
         110
         102
          80
```

```
80
68
120
119
100
85
15...
```

```
80
68
```

16 digits are repeating after decimal.

Q6.

sol: Prime factorisation ~~sho~~ of denominator should contain 2 or 5 or both as factors then the decimal expansion of the number will be terminating.

Q7.

sol: -

1st no. = 1.01001000100001....

2nd no. = 2.02002 000200002....

3rd no. = 3.01001 0001 00001....

Q9.

a.) $\sqrt{23}$ - Irrational

b.) $\sqrt{225} = 15$ = Rational

c.) 0.3796 = Rational

d.) 7.478478... = Rational

e.) 1.1010010001.... = Irrational

Extra ques. - [Q. write in $\frac{p}{q}$ form)

1.) $12.\overline{59}$

$\Rightarrow \frac{1259-12}{99} = \frac{1247}{99}$

2.) $18.5\overline{7}$

$\Rightarrow \frac{1857-185}{90} = \frac{1672}{90}$

3.) $0.00\overline{32}$

$\Rightarrow \frac{32-000}{9900} = \frac{32}{9900}$

4.) $0.253253...$ $(0.\overline{253})$

$\Rightarrow \frac{253-0}{999} = \frac{253}{999}$

5.) $1.\overline{2} + 3.7\overline{6}$

1st + 2nd

$\Rightarrow \frac{12-1}{9}$ $\Rightarrow \frac{376-37}{90}$

$\Rightarrow \frac{11}{9}$ $\Rightarrow \frac{339}{90}$

Adding 1 and 2

$\frac{11}{9} + \frac{339}{90}$

$\Rightarrow \frac{110+339}{90} = \frac{449}{90}$

6.) $0.\overline{6} + 0.\overline{7} + 0.4\overline{7}$

1st + 2nd + 3rd

$\Rightarrow \frac{6-0}{9}$ $\Rightarrow \frac{7-0}{9}$ $\Rightarrow \frac{47-04}{90}$

$\Rightarrow \frac{6}{9}$ $\Rightarrow \frac{7}{9}$ $\Rightarrow \frac{43}{90}$

Adding 1, 2 and 3

$\Rightarrow \frac{6}{9} + \frac{7}{9} + \frac{43}{90}$

$\Rightarrow \frac{60+70+43}{90} = \frac{173}{90}$

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* Finding numbers between any two numbers.

Case I:

when two endpoints (Rational) numbers) & between Rational no.

Method - I

Mean Method (Imp)

=) Mean method → $\frac{\text{sum of all observations}}{\text{no. of observation}}$

Eg-1 → Find 3 rational no's between 4 and 5

1) → $\frac{4+5}{2} = \frac{9}{2}$

2) → $\frac{\frac{9}{2}+4}{2} = \frac{17}{2}$

3.) $\frac{\frac{17}{2}+5}{2} = \frac{19}{4}$

Eg-2. $\frac{1}{2}$ and $\frac{6}{5}$ (2)

1.) $\frac{\frac{1}{2}+\frac{6}{5}}{2} = \frac{5+12}{10} = \frac{\frac{17}{10}}{2} = \frac{17}{10} \times \frac{1}{2} = \frac{17}{20}$

2.) $\frac{\frac{17}{20}+\frac{6}{5}}{2} \Rightarrow \frac{\frac{17+24}{20}}{2} = \frac{\frac{41}{20}}{2} = \frac{41}{20} \times \frac{1}{2} = \frac{41}{40}$

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Note: no. of rational number +1 multiply both no. by this N^r and D^r → then increment the no. by 1 your answer is ready

Method - 2

• => $2 \times \frac{6}{6} = \frac{12}{6}$

=> $3 \times \frac{6}{6} = \frac{18}{6}$

=> $\frac{13}{6}, \frac{14}{6}, \frac{15}{6}, \frac{16}{6}, \frac{17}{6}$ (5 rational no.)

• => $2 \times \frac{10}{10} = \frac{20}{10}$

=> $3 \times \frac{10}{10} = \frac{30}{10}$

=> $\frac{21}{10}, \frac{22}{10}, \frac{23}{10}, \frac{24}{10}, \frac{25}{10}$ .....

Ques. Find 4 rational no.'s between -2 and -5

sol: • $-2 \times \frac{5}{5} = \frac{-12}{5}$

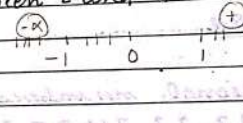
• $-5 \times \frac{5}{6} = \frac{-30}{5}$

=> $\frac{-13}{5}, \frac{-14}{5}, \frac{-15}{5}, \frac{-16}{5}$ ....

-5 ..... -2

Ques. Find 3 rational no. between 2 and -1.

=) 1, 0, ~~2~~, 1.1



Ques- $\frac{1}{2}$ and $\frac{1}{3}$

=) 0.5 and $0.\bar{3}$

=) 0.333 - 0.5

[0.4, 0.41, 0.41]

Case 2 Finding irrational no. between 2 rational no. (between 1 and 2)

1. → 1.101001000 1....

2. → 1.2010010001....

3 → 1.3030030003...

Eg → $\frac{6}{5}$ and $\frac{4}{3}$

- 1.2 - $1.\overline{3}$
- 1.201001000 1...
- 1.211001001000 1....
- 1.202002000 2...

Case - 3 · Find Rational between 2 irrational.

Eg. $\sqrt{2}$ and $\sqrt{3}$

1.414 1.732

- 1.5
- 1.51
- 1.52

Homework

EX - 1.1

Q2) • rational numbers between 3 and 4 (6)

Ans - 3.1, 3.2, 3.3, 3.4, 3.5, 3.6 ...

Q3) Five rational no. between $\frac{3}{5}$ and $\frac{4}{5}$

Ans - 0.6 ⇒ $\frac{3}{5}$, 0.8 → $\frac{4}{5}$ (Ans)

0.60 - 0.80

⇒ 0.61, 0.62, 0.63, 0.64, 0.65, 0.66...

EX - 1.3

Q2. Find 3 irrational no. between $\frac{5}{7}$ and $\frac{9}{11}$

sol.

- $\frac{5}{7}$ → 0.714285 714285.... $(0.\overline{714285})$ (0.71)
- $\frac{9}{11}$ → 0.818181.... $(0.\overline{81})$ (0.81)

irrational no →

1. → 0.7201001000 1....
2. → 0.7301001000 1...
3. → 0.7401001000 1....

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Q. Find 5 rational no. between −2 and 1.

Ans-

- −1
- 0
- 0.1
- 0.2
- 0.3

$\sqrt{2} = 1.414$

	1.414
	$1\,\overline{2.00\,00\,00}$
+1	1
24	100
4	96
281	400
+1	281
2824	11900
+4	11296

$\sqrt{3} =$

	1.732
1	$\overline{3}.\overline{00}\,\overline{00}\,\overline{00}$
+1	1
27	200
+7	189
343	1100
+3	1029
3462	7100
+2	6924

$\overline{11}.\overline{23}\,\overline{45}\,\overline{60}$

LHS — Bar

RHS — Bar Pair

can be on one digit

$\sqrt{5} = 2.236$
$= 2.24$

	2.236
2	$\overline{5}.\overline{00}\,\overline{00}\,\overline{00}$
+2	4
42	100
+2	84
443	1600
+3	1329
4466	27100
+6	26796

Home-work

1. Find 3 rational no. between $\sqrt{5}$ and $\sqrt{16}$.
2. Find 3 irrational no. between $\sqrt{2}$ and $\sqrt{5}$
3. Find 4 rational and irrational no. between $\sqrt{2}$ and 3.
4. Find $\sqrt{24}$, $\sqrt{23}$, $\sqrt{52}$, $\sqrt{13}$ correct upto 2 decimal places.